



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING

DEPARTMENT OF INFORMATION AND COMMUNICATION ENGINEERING

B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: ICT 511 **COURSE TITLE:** Random Process and Queuing Theory

UNITS: 2

TIME ALLOWED: 3 Hours

Instruction: Answer ALL QUESTIONS in Section A and ONE QUESTION EACH from Section B, C and D.

SECTION A

- Which of the following is not a valid probability value?
(A) 0.5 (B) 1.0 (C) -0.2 (D) 0.0
- The CDF $F_X(x)$ of a random variable X must satisfy:
(A) $\lim_{x \rightarrow -\infty} F_X(x) = 1$ (B) $F_X(x)$ is left-continuous (C) $F_X(x)$ is non-decreasing (D) $F_X(x)$ can take values > 1
- For two independent events A and B , which is true?
(A) $P(A \cap B) = P(A) + P(B)$ (B) $P(A \cup B) = P(A)P(B)$ (C) $P(A \cap B) = P(A)P(B)$ (D) $P(A|B) = 1 - P(A)$
- The Poisson distribution is used to model: (A) Continuous measurements (B) Rare event counts (C) Binary outcomes (D) Normally distributed data
- Which set operation represents "A or B occurs"? (A) $(A \cap B)$ (B) $(A \cup B)$ (C) (A^c) (D) $(A \setminus B)$
- Bayes' Theorem updates: (A) Prior probabilities using evidence (B) The sample space (C) Independence of events (D) The mean of a distribution
- The 68-95-99.7 rule applies to: (A) Uniform distributions (B) Poisson distributions (C) Normal distributions (D) Exponential distributions
- If $P(A) = 0.3$ and $P(B) = 0.4$, and A and B are mutually exclusive, then $P(A \cup B)$ is:
(A) 0.7 (B) 0.12 (C) 0.58 (D) 0.0
- Which describes the Law of Total Probability?
(A) $P(A) = P(A|B)P(B)$ (B) $P(A) = \sum_i P(A|B_i)P(B_i)$ for a partition B_i (C) $P(A \cap B) = P(A)P(B)$ (D) $P(A^c) = 1 + P(A)$
- The variance of X is: (A) $E[X^2]$ (B) $E[X^2] - (E[X])^2$ (C) $\sqrt{E[X]^2}$ (D) $E[X]^2$
- The _____ of two events A and B is the set of outcomes in A or B or both.
- For any event A , $P(A^c) = 1 - \underline{\hspace{2cm}}$.
- The _____ rule states $P(A \cap B) = P(A|B)P(B)$.
- A CDF must be _____-continuous.
- Two events are _____ if $P(A \cup B) = P(A)P(B)$.
- The _____ probability of an event is its probability before observing new evidence.
- The _____ function gives $P(X = x)$ for a discrete random variable.
- For a Poisson RV with rate λ , the mean and variance are both _____.
- If A and B are mutually exclusive, then $A \cap B = \underline{\hspace{2cm}}$.
- The _____ of a set S is the collection of all possible outcomes.
- True/False:** The PDF of a single value of continuous RV is always equals to 1.
- True/False:** For any events A and B , $P(A \cup B) = P(A) + P(B)$.
- True/False:** The CDF of a continuous RV is a smooth curve.
- True/False:** $P(A|B) = P(B|A)$ for any events A and B .
- True/False:** The empirical probability is based on theoretical models.

SECTION B

1. (a). (i). Define the Poisson distribution and state its probability mass function (PMF). (2 marks)
 (ii). Name two real-world scenarios where Poisson distribution is applicable. (2 marks)
 (iii). What is the relationship between the Poisson parameter λ and its mean/variance? (2 marks)
 (b). (i). Define the CDF $F_X(x)$ of a random variable X . What are its key properties? (3 marks)
 (ii). How does the CDF of a discrete variable differ from that of a continuous variable? (2 marks)
 (iii). The lifetime of a machine part has CDF $F_T(t) = 1 - e^{-t/100}$. What does $F_T(50)$ represent? (2 marks)
 (c). The Nigerian Navy is conducting a program to update the memory in computers onboard ships. The Navy procures memory modules from three American manufacturers: A1, A2, and A3. The probability of purchasing from each manufacturer is: $P(A1) = 1/6$, $P(A2) = 1/3$, and $P(A3) = 1/2$. Each manufacturer has a different probability of failure for their memory modules: $P(F|A1) = 0.006$, $P(F|A2) = 0.015$ and $P(F|A3) = 0.02$. A technician installs a memory module, and it fails.
 (i). What is the probability that the failed module was from manufacturer A1? (4 marks)
 (ii). What is the probability that the failed module was from manufacturer A3? (3 marks)
 (iii). Based on your results, which manufacturer is the most responsible for failed modules, and why? (4 marks)
2. (a). (i). Write the PDF of a normal distribution $N(\mu, \sigma^2)$ and explain the roles of μ and σ . (4 marks)
 (ii). Why is the normal distribution called the "bell curve"? (3 marks)
 (iii). State the 68-95-99.7 rule and its significance. (3 marks)
 (b). (i). What is a PMF? How does it differ from a PDF? (2 marks)
 (ii). Give an example of a discrete random variable and its PMF. (2 marks)
 (iii). In a gene mutation study, X counts mutations per cell with PMF $P(X = k) = \frac{e^{-2.2} 2.2^k}{k!}$. What distribution is this? (2 marks)
 (c). Simon's Surplus Warehouse sells electronic components in bulk, where each component can be classified as a Resistor (R), Capacitor (C), or Diode (D), and each can be either Good (G) or Faulty (F). Based on historical data, the following probabilities are known: $\Pr[F]$ (Probability of a faulty component) = 0.29, $\Pr[GR]$ (Probability of a good resistor) = 0.03, $\Pr[FR]$ (Probability of a faulty resistor) = 0.12, $\Pr[FC]$ (Probability of a faulty capacitor) = 0.05, $\Pr[FD]$ (Probability of a faulty diode) = 0.32. Assume that every component is exactly one of {R, C, D} and either G or F.
 (i). What is the probability that a randomly selected component is a resistor (either good or faulty)? (2 marks)
 (ii). What is the probability that a component is either faulty or a resistor? (3 marks)
 (iii). What is the probability that a component is useful, i.e., not faulty and not a resistor? (4 marks)

SECTION C

3. (a). A call center receives an average of 5 calls per minute.
 (i). Calculate the probability of receiving exactly 3 calls in a minute. (3 marks)
 (ii). Why might the Poisson assumption break down during a holiday sale? (4 marks)
 (b). (i). If $P(A|B) = P(A)$, what does this imply about events A and B? (2 marks)
 A class has 60% girls and 40% boys. 30% of girls and 50% of boys play chess.
 (ii). What is the probability a randomly selected student plays chess? (4 marks)
 (iii). If a chess player is selected, what's the probability they're a girl? (2 marks)
 (c). A discrete random variable X has the following probability mass function (pmf):

$$P(X=0) = 0.6$$

$$P(X=1) = 0.4$$

$$P(X=2) = 0.0$$

$$P(X=3) = 0.3$$

$$P(X=4) = 0.5$$

$$p_X(x) = \begin{cases} p_X(1) = \frac{1}{2} \\ p_X(2) = \frac{1}{4} \\ p_X(3) = \frac{1}{8} \\ p_X(4) = \frac{1}{8} \end{cases}$$

Bayes' theorem
 $P(A|B), P(B)$
 $P(A|B) \cdot P(B) + P(A|B) \cdot P(B)$

- (i). Plot the pmf of the given function. (2 marks)
 (ii). Find and sketch the cumulative distribution function (CDF) $F_X(x)$. (6 marks)
 (ii). Compute the following probabilities: (a). $P(X \leq 1)$, (b). $P(1 < X \leq 3)$, (c). $P(1 \leq X \leq 3)$. (2 marks)

4. (a). The heights of adults in a town are normally distributed with $\mu = 170$ and $\sigma = 10$ cm. (3 marks)
 (i). What height range covers the middle 95% of the population? (3 marks)
 (ii). Is the normal model valid if the town has many basketball players? (4 marks)
 (b). (i). State Bayes' Theorem and explain each term. (3 marks)
 (ii). In a medical test, why is $P(\text{Disease}|\text{Positive})$ often different from $P(\text{Positive}|\text{Disease})$? (3 marks)
 (iii). How does prior probability $P(A)$ influence the posterior $P(A|B)$? (3 marks)
 (c). The cumulative distribution function (CDF) of the time T (in minutes) it takes a bank teller to serve a customer is defined by:

$$F_T(t) = \begin{cases} 0 & t < 2 \\ A(t-2) & 2 \leq t < 6 \\ 1 & t \geq 6 \end{cases}$$

- (i). Determine the value of A that makes $F_T(t)$ a valid CDF. (3 marks)
 (ii). Using the value of A from part (a), compute $P(T > 4)$. (3 marks)
 (iii). Using the same A , find $P(3 \leq T \leq 5)$. (3 marks)

SECTION D

5. (a). (i). A fair six-sided die is rolled. What is the probability of rolling a prime number (2, 3, or 5)? Why might classical probability fail if the die is weighted? (2 marks)
 (ii). In a clinical trial, 150 out of 500 patients responded to a new drug. Calculate the empirical probability of a patient responding to the drug. If the trial is repeated with 1,000 patients, would you expect the empirical probability to change? Why? (2 marks)
 (iii). For events A and B in a sample space: $P(A) = 0.6$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$. Verify if the axiom of additivity holds for $P(A \cup B)$. If $P(A \cup B) = 0.9$, does this violate any probability axioms? Explain. (3 marks)
 (b). The number of telephone calls arriving at a switchboard during any 10-minute period follows a Poisson distribution with rate parameter $\lambda = 2$. (4 marks)
 (i). What is the probability that more than three calls will arrive during a 10-minute period? (1 mark)
 (ii). What is the probability that no calls will arrive during a 10-minute period? (4 marks)
 (iii). How does $P(X > 3)$ change if λ increases from 2 to 3? Interpret the result. (4 marks)
 (c). The cumulative distribution function (CDF) of the lifetime (in years) of a piece of equipment is given by:

$$f_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x/5} & x \geq 0 \end{cases}$$

- (i). What is the expected lifetime of the equipment? (3 marks)
 (ii). What is the variance of the lifetime of the equipment? (4 marks)
 (iii). Find the probability that the equipment lasts more than 10 years. (2 marks)

6. (a). An insurance company wants to assess the risk of car accidents for drivers aged 18–25 in a city. They use two approaches:

Empirical Probability: Use past accident data from 10,000 drivers in this age group (1,200 had accidents last year).

Axiomatic Probability: Model accidents using a theoretical risk function $P(A) = 1 - e^{-0.15}$ (derived from traffic laws and vehicle safety standards).

Use the above information to answer the following questions

- (i). Calculate the accident probability for each method. (2 marks)
- (ii). Compare the results. Which method is most reliable? Why? (2 marks)
- (iii). Is there a discrepancy in the values of the empirical and axiomatic values? propose reasons for the discrepancy if there is. (2 marks)
- (b). A factory produces light bulbs with lifespans (in hours) that follow a normal distribution with a mean (μ) of 1,200 hours and a standard deviation (σ) of 150 hours. The quality control team needs to answer the following:
- (i). What percentage of light bulbs will last at least 1,000 hours? (3 marks)
- (ii). The factory wants to offer a warranty guaranteeing a minimum lifespan. If they want no more than 5% of bulbs to fail before the warranty period, what should the warranty duration be (in hours)? (2 marks)
- (iii). In a batch of 10,000 bulbs, how many are expected to last between 1,050 and 1,350 hours?
(from standard normal tables: $\Phi(-1.33) = 0.0918$; $\Phi(-1.645) = 0.05$. $\Phi(1.00) = 0.8413$, $\Phi(-1.00) = 0.1587$) (3 marks)
- (c). A student models the lifetime X (in arbitrary units) of a company's equipment using a random variable with the probability density function (PDF):

$$f_X(x) = \begin{cases} \frac{xe^{-x/10}}{100} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- (i). Show that $f_X(x)$ is a valid PDF. (4 marks)
- (ii). Find the probability that the equipment's lifetime exceeds 20. (3 marks)
- (iii). Calculate the expected lifetime $E[X]$. (4 marks)